

Practical 10

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# 1. Load the dataset (using Seaborn's built-in version for convenience)
df = sns.load_dataset('iris')

# 2. Inference: How many features and their types?
print("--- Dataset Information ---")
print(df.info())

# Manual Inference for your journal:
# - Total Features: 5
# - Numeric Features: sepal_length, sepal_width, petal_length, petal_width (Float64)
# - Nominal/Categorical Feature: species (Object/String)

# Setting up the plot area (2x2 grid)
fig, axes = plt.subplots(2, 2, figsize=(12, 10))

# Plotting histograms
sns.histplot(df['sepal_length'], kde=True, ax=axes[0, 0], color='skyblue').set_title('Sepal Length Distribution')
sns.histplot(df['sepal_width'], kde=True, ax=axes[0, 1], color='salmon').set_title('Sepal Width Distribution')
sns.histplot(df['petal_length'], kde=True, ax=axes[1, 0], color='green').set_title('Petal Length Distribution')
sns.histplot(df['petal_width'], kde=True, ax=axes[1, 1], color='orange').set_title('Petal Width Distribution')
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# Setting up the plot area
```

```
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
```

```
# Plotting boxplots
```

```
sns.boxplot(y=df['sepal_length'], ax=axes[0, 0], color='skyblue').set_title('Sepal Length  
Boxplot')
```

```
sns.boxplot(y=df['sepal_width'], ax=axes[0, 1], color='salmon').set_title('Sepal Width  
Boxplot')
```

```
sns.boxplot(y=df['petal_length'], ax=axes[1, 0], color='green').set_title('Petal Length  
Boxplot')
```

```
sns.boxplot(y=df['petal_width'], ax=axes[1, 1], color='orange').set_title('Petal Width  
Boxplot')
```

```
plt.tight_layout()
```

```
plt.show()
```